

Detection of Voltage Droop-Induced Timing Fault Attacks Due to Hardware Trojans

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Abstract—Recent breakthroughs in heterogeneous integration (HI) using 2.5-D/3-D packaging technology have led to several advances in the semiconductor industry, increasing yield while reducing overall cost and time-to-market. However, the diversification of the HI supply chain has led to several sources of distrust due to the use of black-boxed third-party intellectual property (IP), outsourced fabrication, assembly and test facilities during the design and manufacturing process. We demonstrate the susceptibility of chiplet IPs to timing failure due to voltage droop in the power distribution network (PDN) induced by the insertion of chiplet level ring-oscillator (RO)-based hardware Trojans. We present an end-to-end methodology for design, placement, and insertion of RO-based Trojans in chiplet designs followed by characterizing their contribution to the dynamic voltage droop induced within the on-chip PDN. We quantify this PDN impact on timing paths and develop a systematic method to rank the susceptibility of different data paths toward a voltage droop event. We utilize this presilicon security analysis framework to evaluate voltage droop-based attack susceptibility for a variety of IPs, including some from the CEP benchmark. We also develop a machine learning-guided time-series anomaly detection framework to detect voltage droop-based anomalies on functional workloads running on different benchmarks, demonstrating the effectiveness of an convolutional autoencoders in detecting voltage droop-induced timing anomalies.

Index Terms—2.5D ICs, anomaly Detection, autoencoders, hardware Trojans, runtime security, voltage droop.

I. INTRODUCTION

THE GLOBALIZATION of the integrated circuit (IC) supply chain has led to considerable challenges in intellectual property (IP) security. Of particular concern is the vulnerability to IP theft through attacks, such as counterfeiting, IC overbuilding, netlist reverse-engineering, as well as the vulnerability, to runtime security due to malicious design modifications [1], [2]. Security challenges associated with the integration of chiplets on an interposer can lead to threats

arising from within the system, e.g., through rogue or untrusted chiplets. Threats can also arise from outside the system, e.g., through an untrusted end-user [3]. Attacks may be launched with the objective of either stealing the IP (IP theft) or disrupting the normal functioning of the system [denial of service (DoS)]. For example, untrusted chiplets in a system can deliberately affect the functioning of trusted chiplets by increasing the levels of electromagnetic interference (EMI)-induced voltage fluctuations or affect the power distribution networks (PDNs) by introducing voltage droops through malicious modifications in the design [4]. Similarly, an attacker can steal IP by extracting the point-to-point interconnects across chiplet IPs and observing data transfers between chiplets through physical side-channels, such as data buses on the interposer or scan chains [5], [6].

In this work, we explore the security challenges arising from the integration of chiplets that contain malicious modifications. We first review existing solutions for runtime security and identify their shortcomings [7], [8]. A key shortcoming is the assumption of a secure runtime environment during functional operation, which limits the scope of existing protection schemes to threats arising from within the system and restrict them to solutions implemented at a bus-level, e.g., rogue chiplets issuing malicious instructions to make unauthorized requests to memory addresses [8]. We describe a more comprehensive threat model that considers the relevant attack surfaces applicable for chiplet IPs that may be part of a heterogeneously integrated system. We next demonstrate the susceptibility of such IPs to timing failure due to voltage droop in the PDNs. Based on the proposed threat model, malicious modifications, such as the insertion of ring oscillators (ROs), can be made to chiplets fabricated throughout the untrusted supply chain. These ROs, when triggered, can induce voltage droop in the PDNs across both chiplets as well as the interposer. This can exacerbate the delay of functional paths in the design, potentially leading to timing failure during runtime.

We present an end-to-end vertically integrated EDA toolflow that leverages existing industry tools to support the design, layout, and full-chip simulation of chiplet IPs using the ASAP 7-nm standard cell library. We utilize this toolflow to analyze the voltage-droop induced on the PDN due to the activation of RO-based Trojans inserted in chiplets. We next evaluate the voltage droop susceptibility of functional timing paths by derating each timing path based on the worst-case voltage droop experienced by the design running a functional workload. This framework can be used as a presilicon security verification tool to identify potential functional paths that are

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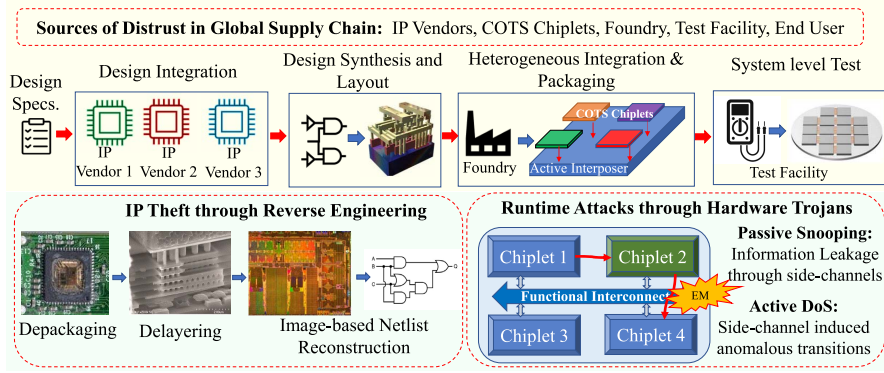


Fig. 1. Sources of distrust arising from the globalized supply chain along with their associated threats: IP vendors, COTS chiplets, untrusted foundries, test facilities, and end users [5].

susceptible to voltage droop-induced timing failures due to the malicious modifications in the supply chain. We also evaluate the sensitivity of the voltage droop anomalies as a function of the number of Trojans inserted in the chiplets of varying sizes. In addition, we develop a machine learning-based time series anomaly detection framework that utilizes autoencoders to detect anomalous voltage droop values triggered due to the activation of RO-based hardware Trojans during runtime. We also investigate the performance of autoencoders in detecting voltage droop anomalies induced by non RO-based Trojans, such as glitch-based power wasting circuits, demonstrating the effectiveness of the ML-based method in detecting voltage droop anomalies for different Trojan types.

The main contributions of this article are as follows: 1) we demonstrate the susceptibility of chiplet IPs to timing failure due to voltage droop in the PDN through insertion of RO-based Trojans and glitch-based power wasting circuits; 2) we develop an end-to-end EDA methodology for design, layout, post-layout voltage droop characterization and timing deration analysis of various chiplet designs with hardware Trojan inserted; 3) we develop a convolutional autoencoder-based time series anomaly detection framework to identify anomalous voltage droop values across chiplet PDNs arising from RO-based and glitch-based hardware Trojans; and 4) we perform a security analysis for different numbers of chiplet-based Trojan insertions in IPs of various sizes.

The remainder of this article is organized as follows: Section II reviews background and prior work. Section III presents the RO-based Trojan design and its characterization. Section IV presents the overall end-to-end flow for voltage droop analysis. Section V presents the security evaluation for timing deration on different IP-level functional workloads. Section VI presents the time series anomaly detection in voltage droop data across different benchmark IPs. Section VII presents a discussion on mitigating voltage droop-based attacks and Section VIII concludes this article.

II. BACKGROUND AND PRIOR WORK

A. Security Threats in Heterogeneous Supply Chain

Emerging threats for heterogeneous integration (HI) systems can be aimed at either IP theft or disruption of functionality during runtime. These threats can be mounted from untrusted

sources that may be internal to the system (untrusted or rogue chiplets) or external in nature (untrusted end users) (Fig. 1).

Rogue or Untrusted IP: Commercial off-the-shelf (COTS) IPs may be sourced from different manufacturers and third-party IP distributors. As a result, without proper trust and authentication, an untrusted chiplet with rogue IP may exercise unauthenticated functions through the shared interconnects implemented on the interposer. Since rogue IPs may have access to the data being transferred through the interconnect, they may intercept, interrupt, modify, or fabricate the data flowing through the common interconnect. Untrusted IPs may include malicious modifications that can cause timing failures due to PDN voltage droop, as demonstrated in this work.

Untrusted End-Users: While rogue chiplets may pose threats that are internal to the functioning of the system, untrusted end users may present threats external to the system. In such a scenario, IP theft is the most prominent threat posed by an untrusted end user because they may try to exploit both physical and scan-based side channels, as well as traditional approaches, such as netlist reverse engineering (RE), to achieve their objectives. Threats arising from rogue or untrusted chiplets are likely to be targeted at disruption of service while those arising from untrusted end users are targeted toward IP theft.

B. Prior Work on IP Security

In [8], a root-of-trust (RoT) system is proposed to protect against system-level threats through the use of a security monitor that implements security policies, also known as hardware security features or HWSFs, that monitor against malicious access of the system's shared memory resources in real time. The protocol consists of policies that manage memory access in a fully controlled setting, ensuring shared memory transactions emanating from trusted and untrusted chiplets are segregated and verified. This protocol is implemented through a secure transaction monitor (TRANSMON) that is also implemented as part of the active interposer. Similarly, another security scheme focuses on achieving runtime security by enforcing system level runtime monitoring of a chiplets through a secure network-on-chip (NoC) architecture, which is implemented in a trusted setting [9]. This NoC runs a protocol called Hybrid Link protocol or HL protocol and supports both point to point and broadcast mode.

Existing RoT-based solutions focus on protection against malicious modifications or system level threats arising from untrusted chiplets or IP theft [5]. This is ensured through the following: 1) a secure interposer that is fabricated in a trusted environment and 2) runtime monitoring protocols that are integrated with the chiplet and enforced by the secure interposer. These security policies are aimed at preventing malicious modification of shared memory and ensure data integrity. However, these methods fail to consider attacks that may be mounted through PDN characteristics such as voltage droops. As such side-channel attacks remain outside the scope of bus-based security solutions, they can manifest as timing failures in functional datapaths of the design. Thus, there exists a need for a side-channel vulnerability assessment framework.

C. Threats Due to Voltage Droop-Based Power Side Channels

Prior work in power-based side channel security has focused on exploiting information leakage through the collection of numerous power traces with the objective of launching differential cryptanalysis attacks against cryptographic IP [10]. Such cryptanalysis-based attacks are targeted to reveal encryption keys. Cryptographic applications have been shown to be easily susceptible to power-based side channel attacks due to correlation between the total power consumed by an IP and the cryptographic workload being run on the IP [11]. The importance of on-chip PDNs in the context of such power-based side channel attacks has also been investigated [12].

The increasing use of secure programmable hardware, including FPGAs, as on-chip security engines has also motivated the analysis of power-based side-channel attacks for programmable logic. Gnad et al. [13] showed voltage droop due to the insertion and activation of ROs as an effective method to induce runtime faults, leading to functional failure of cloud-based FPGAs. In [14], the effectiveness of asynchronous ROs to launch voltage-based timing failures on FPGA-based deep learning accelerators has been demonstrated. Both these attacks exploit RO-based malicious modifications to introduce timing failures. Designing methods have also been studied to mitigate such power-based side channel leakage in FPGA settings. In [15], ROs were repurposed to act as sensors that remain sensitive to changes in the voltage droop. These sensors can be implemented at the PCB level to monitor voltage droop, but they remain susceptible to errors and noise. In [16], an active fence of ROs is inserted between the victim and the attacker IP that creates an additional layer of workload-independent voltage fluctuations, thereby reducing the susceptibility to correlation-based power analysis attacks.

D. Limitations of Prior Work on Voltage Droop-Induced PDN Security

Voltage droop-based attacks and countermeasures for both cryptographic and FPGA-based use-cases have been studied in prior work [17], [18]. Attacks, such as [17], exploit the timing vulnerability of FPGA platforms to on-chip voltage droop events triggered by on-board ROs. While ROs are used to cause timing failure, no countermeasures are proposed in

that work to either detect or prevent such RO-induced timing attacks during runtime.

A technique to detect Trojans that specifically impact circuit path delay has been proposed in [19]. This method uses clock frequency sweeping to generate delay signatures for different logic paths in the design. These signatures can be impacted by a Trojan's load capacitance, thereby allowing designers to identify delay-based Trojans. Vakil et al. [20] extended this work by improving the timing model to account for delay noise and uses a neural network to track process drift. However, both these methods remain workload agnostic and fail to account for situations where dynamic workloads may induce timing failure during runtime. These methods also do not provide a learnable mechanism to track the state of the IP during runtime and detect attacks in functional mode. Individual current-sensing path delay monitors that convert current activity on local power grid into a timing pulse have also been proposed to screen delay-based hardware Trojans [21]. These monitors can be integrated with current comparators calibrated to reduce the impact of process variations and deployed in clusters connected to scan chains. This method relies on the use of scan chains to detect Trojans, thereby preventing the ability to detect them during runtime in functional mode. In [22], an accurate characterization of path delays is performed while also modeling worst-case IR drop in power rails. This IR drop-aware timing characterization is then used to suggest design optimizations to improve circuit power and performance. However, these optimization-based methods [22], [23] in IP design can lead to narrower margins on many timing paths, making such designs even more susceptible to workload-driven voltage droop events. Zeng et al. [24] presented a method to calibrate ROs under the presence of IR-drop for hardware Trojan detection, but it also does not consider dynamic workloads or timing impact on path delays.

The prior methods discussed above also do not perform a Trojan-based sensitivity analysis to deal with scenarios where dynamic loading conditions of the IP may cause Trojans, which originally have minimal impact on path delays, to cause timing failure due to dynamic changes in workload. They also do not include a learnable method to characterize Trojan behavior in dynamic runtime conditions, and therefore have limited applicability for runtime detection.

E. Threat Model

As discussed in Section II-A, there exist multiple sources of distrust in the paradigm of heterogeneously integrated systems. Therefore, the threat model assumes the following: 1) use of black-boxed COTS IPs with limited visibility to the internals of the IP, allowing attackers to insert malicious modifications in the design and 2) use of untrusted foundries and OSATs for chiplet/IP fabrication. Thus, the integrated system can be vulnerable to attacks mounted by either the COTS IP designers, untrusted foundry, or OSAT facilities. Without an effective solution to protect the different IPs in the design, it is impossible to guarantee the security of a heterogeneous system, given that many of its components are sourced from untrusted entities.

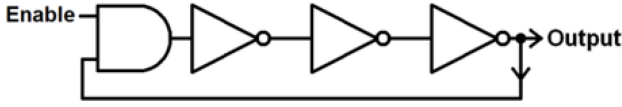


Fig. 2. Example RO design that can be inserted in a malicious chiplet.

Exhaustive security verification of black-box IPs is challenging; it might not always be possible for system integrators to visually inspect and verify such blocks. Thus, it is possible for untrusted users to make malicious modifications that can then be triggered during functional operation of the design, leading to DoS type attacks. For this study, we focus on modifications made within the same chiplet. Prior work has shown that these modifications can be introduced through foundry side engineering change order (ECO) flows that make layout level modifications within third-party chiplets [25], [26], [27], [28].

III. RO-BASED TROJANS IN FUNCTIONAL IPs

ROs are widely used for a variety of in-silicon applications, including process characterization, aging monitoring, and lifecycle management [29], [30], [31]. The voltage carried across a PDN droops due to variable loading characteristics across the PDN. The voltage droop V_{droop} across a power line is proportional to the resistance of the line along with the inductive impedance across the same line. Inductive effects arise from rapid differences in the electrical current and is influenced by both spatial and temporal circuit switching activity, which in turn is dependent on system behavior. Thus, $V_{\text{droop}} = IR + L(di/dt)$.

ROs, when triggered, act as fast switching components drawing significant amount of current and inducing voltage droop in the power rails for downstream logic. This can exacerbate the delay of timing critical paths in the design. This voltage droop induced timing deration is quantified as follows [32]:

$$T_{\text{droop}} = T_{\text{nom}} \times \frac{1 - \frac{V_{\text{th}}}{V_{\text{nom}}}}{1 - \frac{V_{\text{th}}}{(V_{\text{nom}} - V_{\text{droop}})}} \quad (1)$$

where V_{nom} is the nominal supply voltage and T_{nom} is the path delay without any voltage droop. For the ASAP 7-nm process, $V_{\text{nom}} = 1.1$ V and $V_{\text{th}} = 0.15$ V.

A. Characterizing RO-Induced Voltage Droop

In this section, we describe the design and architecture of an example RO-based Trojan that can be potentially inserted into a chiplet or the interposer. Fig. 2 illustrates the design of a 3-stage RO. Each RO is triggered by an AND gate that is driven by an enable signal. The standard cells used for the RO design share the same PDN as the functional logic within the chiplet (and the interposer if it is of the active type), thereby inducing voltage droop for the functional logic downstream. Fig. 3 illustrates the layout (a) and voltage droop distribution (b) for every standard cell in a 31-inverter RO. Each inverter in the RO chain has a total internal power consumption of about 49.3 mW, with the 31-inverter RO occupying a footprint of 39.09 μm^2 . The small footprint makes it easy to insert such ROs in black-boxed chiplet IPs as well as insertion within the

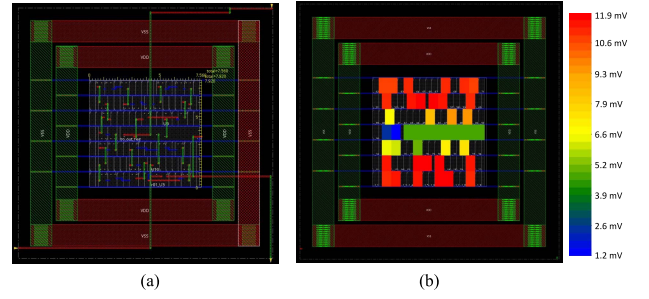


Fig. 3. (a) Illustration of layout for a 31-stage RO with mesh-type PDN and (b) distribution of V_{droop} for the same RO across a range of values from 1.2 mV (blue) to 11.9 mV (orange).

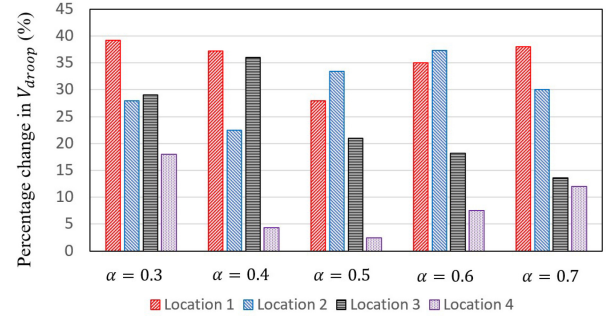


Fig. 4. Percentage change in voltage droop at four different locations within the FIR chiplet due to a 31-stage RO, evaluated across four different switching activities (denoted by α) of the IP.

untrusted interposer. Existing work considers the interposer to be the hardware RoT, assuming it to be fabricated in a trusted environment. However, we consider the most conservative threat model, which includes the design and fabrication of both chiplets and the interposer in untrusted environments, thereby making it easy for attackers to place such RO-based modifications within the design.

The ROs inserted in the design can impact the voltage droop dynamically. As an illustration, we evaluate the impact of a 31-stage RO on the worst-case voltage droop when the RO is inserted in a single chiplet implementing the FIR filter IP from the common evaluation platform (CEP) benchmark suite [33] as illustrated in Fig. 4. We insert the RO at the center of the FIR chiplet and evaluate the change in the worst-case voltage droop across four different locations in the FIR chiplet through static power analysis. In static power analysis, we vary the switching activity, also known as the activity factor (α), of the inputs and sequential elements in the design. This evaluation is done for different values of activity factor and meant to simulate a variety of loading conditions on the IP. Higher the value of α , greater is the switching activity of the IP. The input and sequential activity factors are together varied from 0.3 and 0.7. The four locations are picked from the standard cells located at the center of the four quadrants of the FIR chiplet, when it is partitioned in equal halves vertically and horizontally across the center. Details of the evaluation methodology are presented in the next section.

RO Sensitivity: Note that the voltage droop value can vary for the same location depending on the dynamic loading conditions imposed on the system due to change in inputs over time. Thus, it is important to analyze variations in the voltage droop across a variety of system specific switching

activity. This implies that at any given time, the functional state of the IP determines the sensitivity of the voltage droop. Similarly, for a given state of the IP, different locations in the IP are susceptible to different values of voltage droop. Thus, V_{droop} is a function of the input state of the system as well as the location. Thus, the presilicon evaluation of the worst-case voltage droop can help in evaluating likely regions or hotspots most susceptible to timing failure.

IV. VOLTAGE DROOP-BASED PDN ANALYSIS

In this section, we describe how existing RTL-to-GDSII flows (using standard industry EDA tools, such as Synopsys design compiler and Cadence Innovus) can be utilized to perform timing-aware voltage droop analysis. The overall methodology that we have developed can perform design, layout, and post-layout voltage droop and timing analysis for chiplet IPs that could be later integrated in 2.5-D/3-D systems.

A. Tool Specific Layout Details

The system-level RTL is synthesized using Synopsys DC. We rely on the ASAP 7-nm standard cell library for our RTL-to-GDSII flow. The different chiplet IPs can be either black-boxed COTS IPs or designed in house. The chiplets are then passed through the floorplanning stage followed by micro-bump assignment, place, and route; all done using Cadence Innovus. All designs are based on the ASAP 7-nm PDK. The μ -bump pitch considered is 20 μm . The bottom six metal layers of the ASAP back-end-of-line (BEOL) stack are used for intrachiplet routing to create hard chiplet macros. The μ -bump arrays for the chiplets are placed on metal-7 ($M7$) as pin constraints for interposer routing.

B. Chiplet PDN Design and Voltage Droop Analysis

PDN Architecture: Chiplet PDN is implemented through the insertion of V_{DD} and V_{SS} stripes. Sets of V_{DD} and V_{SS} lines are placed in an alternating fashion with a fixed spacing between the two. The power stripes are inserted in metal layers $M5$ and $M6$. The set-to-set distance between each set of alternating V_{DD} and V_{SS} lines is also predefined as 5 μm for both $M6$ and $M7$. Stripes along $M6$ run in the X-direction while stripes along $M5$ run in the Y-direction. V_{DD} and V_{SS} rings enclose each chiplet.

Fig. 5 illustrates the PDN insertion flow using Cadence Innovus. Power rings can also be inserted in such designs. Following PDN insertion, the chiplet level routing is performed that generates the final outputs.

Once the PDN insertion is completed and the final GDSII files are generated, the voltage droop analysis workflow can be invoked through Cadence Voltus, which supports both static and dynamic power analysis. Our flow supports voltage droop evaluation through *static power analysis (vector-less)* using activity factors (α) that specify switching activity of sequential cells and inputs, that can range between 0.1 and 0.9. Our flow also supports voltage droop evaluation through *dynamic power analysis (vector-based)* using value change dump (VCD) files that are generated from functional Verilog testbenches for the chiplet designs being evaluated. The chiplet is first designed to operate on a particular clock frequency which

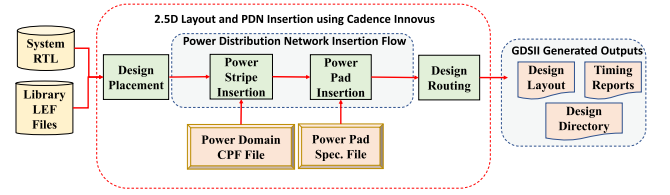


Fig. 5. PDN insertion flow using Cadence Innovus.

is closest to its maximum possible operational frequency, called f_{max} . Static power analysis only requires specifying the switching activity for both inputs and sequential cells, respectively, ranging between 0.1 and 0.9. Dynamic analysis needs a functional workload that is specified within the Verilog testbench customized for an individual chiplet based on its functional specifications.

C. Case Study—FIR Filter Chiplet

We utilize the above tool flow to evaluate the voltage droop for the FIR chiplet through static analysis. The FIR chiplet consists of a 32-bit digital FIR filter that has been obtained from the CEP benchmark suite [33]. We use the ASAP 7-nm library for the flow. Note that ASAP7 is a predictive PDK and implemented designs may contain DRC errors specific to the libraries used. A list of such DRC errors and justification for their waivers is provided in [34]. This has no impact on voltage droop analysis. The chiplet aspect ratio is set at 0.7 which is very close to 1. This allows for a more uniform distribution of microbumps, which are responsible for connecting the input and output pins of the chiplet to the interposer, across the entire area of the chiplet. Power stripes are also added along both horizontal and vertical directions. A single pair of power pins are inserted at the center of the design, for V_{DD} and V_{SS} , respectively. No routing constraints are placed other than microbump insertion at $M7$.

When Voltus is executed, the chiplet-level voltage droop is evaluated for a full range of input activity factors. Fig. 6 shows the step-by-step breakdown of the entire flow for chiplet-level voltage droop evaluation with respect to the FIR chiplet design. Chiplet-level voltage droop values are evaluated for four different input activity factors: (0.3, 0.4, 0.5, 0.6, and 0.7). Note that blue indicates the lowest-voltage droop while red indicates region of maximum voltage droop. We observe that as the distance from the power pins (blue regions at the center of the chip) increases, the voltage droop values also increase. The worst-case voltage droop across the chip is in the range of 9 mV to 60 mV.

V. SECURITY ANALYSIS—EVALUATING VOLTAGE DROOP-INDUCED TIMING FAILURE

In this section, we present the overall security verification methodology for ranking functional timing paths in the design based on their susceptibility to the worst-case PDN voltage droop through dynamic power analysis. We evaluate the derating factor for each timing-critical path with ROs inserted in the chiplets and present an approach to rank functional timing paths based on the voltage droop impact. Since voltage drooping in chiplet PDNs is a dynamic phenomenon that

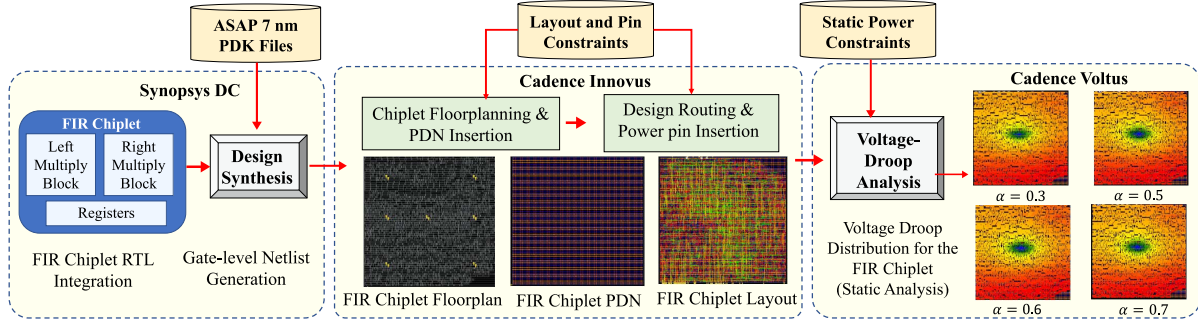


Fig. 6. Illustration of the end-to-end design and voltage droop evaluation toolflow, including the chiplet floorplan output, PDN placement output, chiplet layout output, and voltage droop output across four activity factors for the FIR chiplet.

changes with time depending on the type of workload running on the IP, it is important to perform timing deration analysis of chiplet logic paths while simulating functionally accurate workloads. These functional workloads must be determined based on the chiplet's functional specifications. Therefore, we ensure that the workloads used for timing deration studies using vector-based voltage droop analysis are characterized to represent the functional use-case of the IP and match the type of input vectors the IP will experience during functional operation in-field.

A. Timing Deration Due to Voltage Droop

Let $\mathcal{P}$ denote the set of all timing paths in a chiplet. Let us consider the i th path, $p_i \in \mathcal{P}$, such that its nominal path delay is given by $t_{\text{nom},i}$. For a given workload $\mathcal{W}$ running on the chiplet with an average activity factor of α , the worst-case voltage droop for all standard cells in the path p_i will change the total path delay to $t_{\mathcal{W},i}$ such that $t_{\mathcal{W},i} = t_{\text{nom},i} + t_{\text{droop},\mathcal{W}}$, where $t_{\text{droop},\mathcal{W}}$ is the V_{droop} induced timing deration factor added to the overall delay. Using this approach, we evaluate the timing deration metric for the different paths in the design. Depending on its size and the process node used for fabrication, a chiplet can have a large number of timing paths [35]. For example, design supported by Intel's Foveros 3-D packaging technology may contain hundreds of thousands of paths [36], [37]. As a result, depending on design complexity, security engineers can restrict this evaluation to the N most critical timing paths in the design. Furthermore, functionally critical paths that may not be timing critical but require security coverage (e.g., paths responsible for FSM control logic, paths in fan-out cones of primary outputs [38], [39]), can also be added to this analysis. For every standard cell instance in the path, we obtain the worst-case voltage droop experienced by that instance upon the activation of the RO-based Trojan. This worst-case voltage droop value is obtained through dynamic vector-based analysis while running a functionally characterized workload on the IP. In our analysis, without loss of generality the value of N is set to 100. As the voltage droop data used to evaluate timing degradation is obtained from workload-based post-layout simulation, an accurate estimate of the timing derating factor $t_{\text{droop},\omega}$ can be obtained.

The selection of timing paths to be included for timing deration in $\mathcal{P}$ can be carried out by the designers based

on their security requirements. The total time incurred in calculating path derating factors is directly proportional to the number of instances in each path and the total number of paths in the design. Thus, the total time for calculating the path derating factors is directly proportional to the total number of instances in all the paths in $\mathcal{P}$. Thus, the designers can limit the total number of paths in $\mathcal{P}$ based on the following factors.

- 1) *Slack Threshold*: The minimum value of slack that is deemed necessary for all the paths in the design. This is determined based on the system requirements and operating clock frequency. All paths with slacks lower than the threshold are subject to timing deration analysis.
- 2) *Functional Criticality*: Certain paths in the design might be functionally critical and are subject to timing deration analysis [40].
- 3) *Voltage Droop Distribution*: The distribution of the worst-case voltage droop for the chiplet design can be obtained using the methodology presented in this article. Designers can then identify instances in the layout that experience worst-case voltage droop above a given threshold (say 10% of nominal voltage droop), and paths containing those instances, for timing deration.

B. Security Analysis Methodology

We propose a presilicon security analysis framework/methodology that can generate a ranked list of paths that are most susceptible to timing failure due to voltage droop in the PDN induced through malicious modifications. The security analysis consists of the following steps.

- 1) Generating the layout of the 2-D chiplet IP including PDN.
- 2) Extracting the timing report of all the critical paths through Cadence Tempus.
- 3) Running Trojan-free voltage droop simulation to identify the distribution of worst-case voltage droop.
- 4) Inserting RO blocks within the design and extracting worst-case voltage droop for different workloads (static or dynamic). Note that for dynamic analysis, a workload consists of the application of functionally correct inputs to the chiplet IP based on its system specifications through a Verilog testbench.
- 5) Evaluating timing deration due to worst-case voltage droop for the timing (and security) critical paths.

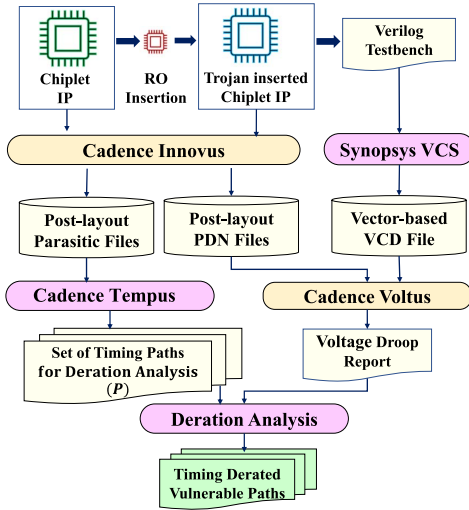


Fig. 7. Methodology for ranking timing-critical paths due to voltage droop through dynamic power analysis.

Parasitic extraction is done on the post-layout netlists using Cadence Innovus, following which Cadence Tempus is used to perform timing analysis. The overall security analysis methodology is illustrated in Fig. 7. We begin with chiplet designs, such as FIR, IIR, or 3DES IPs from the CEP benchmark suite. The chiplets for each of these IPs are designed through Cadence Innovus using the ASAP 7-nm PDK. The gate-level netlists are generated using Synopsys Design Compiler. Following chiplet design, the post-layout parasitics are extracted for running timing analysis using Cadence Tempus, which provides a detailed timing report. ROs are then inserted in the layout and Cadence Voltus is used to obtain the worst-case voltage droop for all the instances in the design using both static (vector-less) and dynamic (vector-based) power analysis. For dynamic power analysis, Synopsys VCS is used to generate the chiplet's VCD file, which is based on its functional Verilog testbench. The VCD file along with the chiplet layout is then used by Voltus to evaluate workload specific voltage droop. Using the voltage droop data, all the timing paths reported through Cadence Tempus are re-evaluated, taking into account the timing derating factor computed from the instance-level voltage droop data. This analysis allows us to analyze the timing degradation due to RO-induced voltage droop for the selected paths in the design.

C. Workload Characterization for Vector-Based Voltage Droop Analysis

As described earlier, it is essential to match the vector-based inputs (used during dynamic voltage droop analysis) as closely as possible to the type of inputs experienced by the IP during functional operation. As a result, the workload for an IP X , represented as $\mathcal{W}_X$, is characterized by a sequence of l input patterns, such that $\mathcal{W}_X = [I_1, I_2, \dots, I_l]$. Here, I_j corresponds to the j th input pattern/vector in the workload or sequence $\mathcal{W}_X$. Since every sample I_j in the workload $\mathcal{W}_X$ is applied consecutively, $\mathcal{W}_X$ can also be considered to be time series data. The size or width of the input pattern/vector I_j is determined by the size of the primary inputs associated with the IP X . Note that clock and reset signals are not

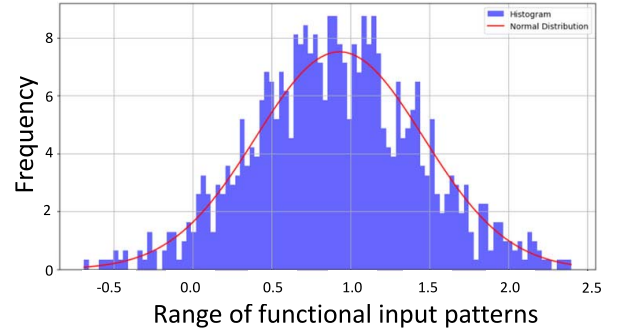


Fig. 8. Gaussian distribution of the sample space of all functional input patterns associated with the FIR filter.

considered to be a part of I_j as they are global signals for any IP. Furthermore, the length of the time series workload or input sequence, l , can be set depending on the number of input samples required for time series data analysis. For our experiments, $l = 1000$, i.e., every workload consists of a series of 1000 samples. Dynamic voltage droop analysis can now be run any window of consecutive input samples in $\mathcal{W}_X$ to obtain the worst-case voltage droop value due to switching activity caused by inputs in that window. The window size, Δ , can vary from 2 (worst-case voltage droop between two consecutive inputs patterns) to l (worst-case voltage droop across all input patterns in $\mathcal{W}_X$). Note that time series voltage droop values can be obtained for an IP X , running a workload $\mathcal{W}_X$, by setting $\Delta = 2$. This is given by $\mathcal{V}_X = [V_{1-2}, V_{2-3}, V_{3-4}, \dots, V_{l-1-l}]$, where V_{1-2} is the worst-case voltage droop value between I_1 and I_2 . Time series voltage droop data, $\mathcal{V}_X$, will be used for runtime anomaly detection described later in Section VI. For timing deration analysis, we utilize the worst-case voltage droop across the entire workload. Therefore, $\Delta = l$ for timing deration analysis. This voltage droop data can be obtained for every cell instance in a chiplet logic path.

To ensure that the workload used for dynamic V_{droop} analysis is as close to functional in-field specification for the IP, we further characterize $\mathcal{W}_X$ into two modes of operation, busy mode ($\mathcal{W}_{X,\text{busy}}$) and *bursty mode* ($\mathcal{W}_{X,\text{bursty}}$). In busy mode, the IP receives input transactions (input patterns) consecutively with no idle time in between. Every input, I_j in $\mathcal{W}_{X,\text{busy}}$, is sampled from a Gaussian distribution of all possible input samples. In bursty mode, the inputs patterns sampled from the same Gaussian distribution are interspersed with periods of low-switching activity ($\alpha = 0.1$). As a result, depending on the burst factor, $\beta \in [0, 1]$, the workload $\mathcal{W}_{X,\text{bursty}}$ can have a fraction of inputs (given by β) sampled from a Gaussian distribution with the other inputs having α of 0.1. Fig. 8 illustrates the distribution of functional input patterns associated with the FIR filter, which is part of the CEP benchmark suite. Note that X-axis represents the decimal value associated with the 32-bit input vector for the FIR filter.

D. Results and Evaluation

1) *Benchmark Setup*: We illustrate this methodology on chiplet designs obtained from the CEP and OpenCores benchmark suites. We utilize the chiplet design flow presented in Section IV to design five different chiplets, namely, the FIR

filter, IIR filter, 3DES, MD5, and JPEG encoding chiplet IPs. All chiplet designs being analyzed range from small to large sized IPs with the JPEG encoder being the largest design with an area of $1,546,240 \mu\text{m}^2$, the MD5 benchmark having an area of $217,800 \mu\text{m}^2$, the FIR filter having an area of $5318 \mu\text{m}^2$, the IIR filter having an area of $8812 \mu\text{m}^2$, and the 3DES IP being the smallest at about $3272 \mu\text{m}^2$. Table I shows the relative size of every chiplet in terms of the number of standard cells, pin count, X and Y dimensions, and total area. We clarify that for voltage droop analysis, the active chiplet area is enclosed by power and ground rings with multiple PG pins placed along the periphery of the ring.

2) *Evaluation Approach*: The chiplets are designed to run functional workloads (for dynamic power analysis) that are obtained from the Verilog unit simulation testbench supplied with the CEP and OpenCores benchmark suites for IP validation. The Verilog testbench exercises the IP's functional inputs based on the IP specifications. The workloads are designed based on the approach outlined in Section V-C. Two workloads are run, $\mathcal{W}_{X,\text{busy}}$ and $\mathcal{W}_{X,\text{bursty}}$. Note that all the designs are run at their corresponding f_{max} frequencies. Based on the sampling of the workloads, the stimulus files for the testbenches are generated in Verilog. The DSP chiplets (FIR and IIR filters) accept input data packets to perform FIR and IIR-based cross correlation operations while the 3DES chiplet computes the ciphertext, respectively, for a set of plaintext input patterns. RO-based Trojans are also inserted with their enable pins connected to the system-level reset pin such that the ROs are triggered whenever the reset pin is deasserted, ensuring that the Trojans are activated as soon as the design is brought out of the reset state. As the voltage droop in the PDN fluctuates with the functional workload, we collect the worst-case voltage droop data through Voltus for both workloads. The voltage droop data obtained through power PDN rail simulations is then used to calculate the derating factors for the N -worst-timing paths in the design (N being 100 for this analysis).

3) *Experimental Results*: We evaluate the average timing degradation for the 100 worst-timing paths, due to different numbers of ROs inserted in the 2.5-D chiplet design. We also evaluate the number of paths that lead to a timing violation due to the ROs inserted in the chiplets. All experiments were conducted on an Intel Xeon Gold CPU running at 2.1 GHz. For the largest benchmark, the average CPU runtime for voltage droop analysis using Cadence Voltus was approximately four minutes. Similarly, timing analysis using Cadence Tempus took less than two minutes. Table II shows the percentage degradation in timing for the 100 worst paths, i.e., paths with least amount of slack, in the different chiplet designs due to the insertion and activation of different number of RO units. We report both the average degradation in the 100 worst-paths, $\mu_{N,\text{worst}}$ and also the standard deviation in the percentage degradation across the 100 worst-paths, $\sigma_{N,\text{worst}}$. The timing impact increases with the number of RO units inserted. This is because increasing the number of ROs in the design also increases the amount of voltage droop in the design.

Table II also presents the number of paths with timing violation as a result of RO activation. We also analyze the impact of the ROs on the worst-affected paths in the design.

TABLE I
CHIPLET SIZES FOR CEP AND OPENCORES EVALUATION BENCHMARKS
IMPLEMENTED USING ASAP7 PDK

IPs	Standard Cells	Pin Count	Active Logic X (μm)	Active Logic Y (μm)	Logic Area (μm^2)
3DES	2018	298	290	160	46,400
FIR	3450	66	360	180	64,800
IIR	4773	66	340	170	57,800
MD5	11037	136	660	330	217,800
JPEG	506578	70	1280	1208	1,546,240

We observe that for certain paths in the design, the timing impact is much more significant, sometimes even leading to timing failure. This is because although a path is not originally timing critical, part of its logic passes through the region experiencing the worst-case voltage droop in the chiplet. As a result, running this analysis for the selected paths in the design provides designers with insights into potential regions on the chip that may be susceptible to timing failures. This is made possible by combining both voltage droop information obtained from the 2.5-D layout along with accurate timing reports obtained using parasitic files extracted post-layout.

E. PDN Voltage Droop Sensitivity for Variable Trojan Counts

In this section, we evaluate the sensitivity of PDN voltage droop on the number of Trojans inserted in every chiplet. We vary the number of RO-based Trojans embedded in every chiplet and analyze the percentage change in voltage droop as a function of Trojan count. We also evaluate the average percentage degradation in the N worst-timing paths due to Trojan insertion for every benchmark. This evaluation is done on a mixed workload, $\mathcal{W}_{X,\text{mix}}$, which consists of equal number of samples taken both $\mathcal{W}_{X,\text{busy}}$ and $\mathcal{W}_{X,\text{bursty}}$ and interspersed uniformly to represent a general functional use case for every chiplet IP. This ensures that a realistic workload setting is maintained for the evaluation of voltage droop-induced timing degradation for every IP. Fig. 9 (Top) shows the increasing trend in PDN voltage droop as the number of embedded Trojans is increased from one to thirty. The bottom figure shows a corresponding increasing trend in the percentage degradation of the N worst-timing paths ($N = 100$) due to Trojan-induced change in V_{droop} . Note that inserting a single RO-based Trojan introduces an average degradation of 0.5%–0.8% in critical timing paths across the chiplets being evaluated. Increasing the Trojan count to three and five increases the percentage degradation to approximately 1.5%. We start observing timing path violations in IIR and FIR IPs at the insertion of five Trojans with three and six violations, respectively. Two and seven violations are observed in 3DES and MD5 chiplets at the insertion of seven Trojans with none before.

Susceptibility of Designs to RO-Based Trojan Insertion: From the discussion above, it is evident that as little as five RO-based Trojans when triggered in sync can cause timing failure in certain chiplets. These RO-induced timing failures can also be exacerbated for designs with narrow timing margins, making such designs more sensitive to even

TABLE II
PERCENTAGE DEGRADATION IN N -WORST-TIMING PATHS DUE TO TROJAN-INDUCED VOLTAGE DROOP TIMING DEGRADATION IS EVALUATED FOR $W_{X,busy}$ AND $W_{X,bursty}$ WHILE RUNNING AT f_{max} , WITH $N = 100$

No. of ROs	$W_{X,busy}$											
	FIR Chiplet			IIR Chiplet			3DES Chiplet			MD5 Chiplet		
	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}
15	3.34	0.40	21	3.04	0.69	13	2.01	0.09	4	3.09	0.59	12
20	3.27	0.43	22	3.78	0.34	16	2.14	0.08	4	4.18	0.48	15
25	3.54	0.49	23	5.18	0.55	28	2.20	0.07	7	4.53	0.43	19
30	4.19	0.43	25	5.42	0.58	30	2.45	0.05	8	5.02	0.53	20
No. of ROs	$W_{X,bursty}$											
	FIR Chiplet			IIR Chiplet			3DES Chiplet			MD5 Chiplet		
	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}	$\mu_{N,worst}$	$\sigma_{N,worst}$	P_{crit}
15	3.39	0.33	22	3.33	0.87	13	1.64	0.11	3	3.12	0.54	12
20	3.45	0.39	23	5.89	0.67	15	1.70	0.09	3	4.23	0.43	16
25	3.75	0.45	23	6.15	0.64	34	1.87	0.11	8	4.68	0.55	20
30	4.19	0.49	25	6.29	0.65	34	2.54	0.87	8	5.05	0.41	20

$\mu_{N,worst}$: Average percentage degradation in the N -worst timing paths of the chiplet; $\sigma_{N,worst}$: Standard deviation in the percentage degradation of the N -worst timing paths of the chiplet; P_{crit} : No. of timing paths violated.

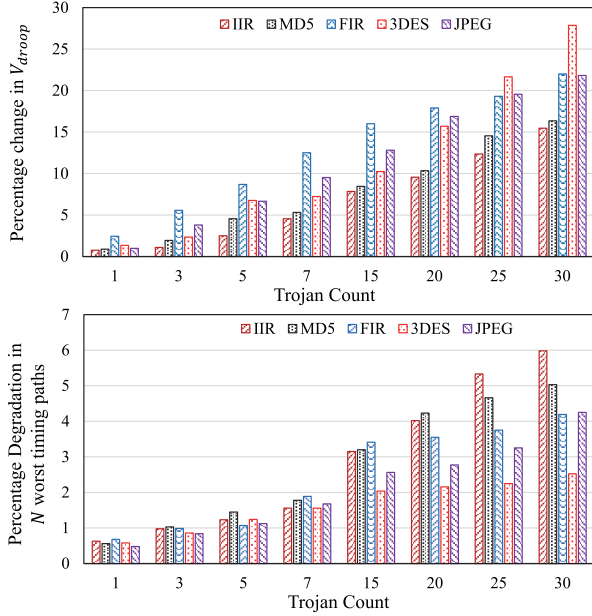


Fig. 9. (Top) Percentage change in V_{droop} and (bottom) percentage degradation in 100 worst-timing paths, as a function of increasing number of Trojans in the design.

a small number of Trojan embeddings. Recent work has shown that there exist several pathways for attackers to insert malicious modifications in the design through the use of ECOs, layout level modifications, and trust-gaps within the supply chain [25], [26], [27]. Furthermore, ROs are widely used in silicon for a variety of applications, such as process characterization, aging monitoring, and Silicon lifecycle management [29], [30], [31]. Due to the prevalence of on-chip ROs, it is easier for attackers in an untrusted supply chain to reconfigure a certain subset of nets and derive a trigger signal from an internal net in the design, making the process of malicious modification for RO-based Trojans easier.

VI. DETECTING VOLTAGE DROOP ATTACKS USING TIME-SERIES ANOMALY DETECTION

In the previous section, we demonstrated the effectiveness of RO-induced voltage droop in causing timing degradation for critical paths in chiplets. In this section, we present an ML-guided method to detect such voltage droop attacks in time series data using convolutional autoencoders [41]. We also highlight our approach to minimize false positives while demonstrating the effectiveness of convolutional autoencoders for time series attack detection in voltage droop data across several benchmarks from the CEP suite.

A. Time Series Anomaly Detection Using Autoencoders

Autoencoders are neural networks that learn to reconstruct the supplied input sample, thereby producing a mapping between the reconstructed input sample and itself [41]. This is achieved by downsampling an input feature vector, x through an encoder architecture, E_θ to generate a latent space vector, z . This is then upsampled through a decoder architecture, D_ϕ to regenerate a feature vector, $\bar{x}$ which is the same size as the input vector, x . The autoencoder can then be trained to minimize the reconstruction loss for an input sample, x_i , such that $\min_{\phi, \theta} \{\mathcal{L}(\phi, \theta; x_i)\}$, where $\mathcal{L}(\phi, \theta; x_i) = (1/N) \sum_{i=1}^N \|x_i - D_\phi(E_\theta(x_i))\|_2^2$. Here, $\|\cdot\|$ is used as the $L2$ norm, N is the number of samples used for training. Autoencoders can learn the distribution of a set of samples on which they are trained. As a result, if a sample x , supplied to the trained autoencoder belongs to the same distribution, the value of $\mathcal{L}(\phi, \theta; x)$ will be very low for that sample. Therefore, an autoencoder can also be used identify anomalous data as the value of $\mathcal{L}(\phi, \theta; x)$ for such samples will be very high. After training an autoencoder to in-distribution data, a reconstruction loss threshold, ϵ can be identified. If $\mathcal{L}(\phi, \theta; x) < \epsilon$, the incoming sample is classified as nonanomalous. However, if $\mathcal{L}(\phi, \theta; x) > \epsilon$, the sample is classified as anomalous.

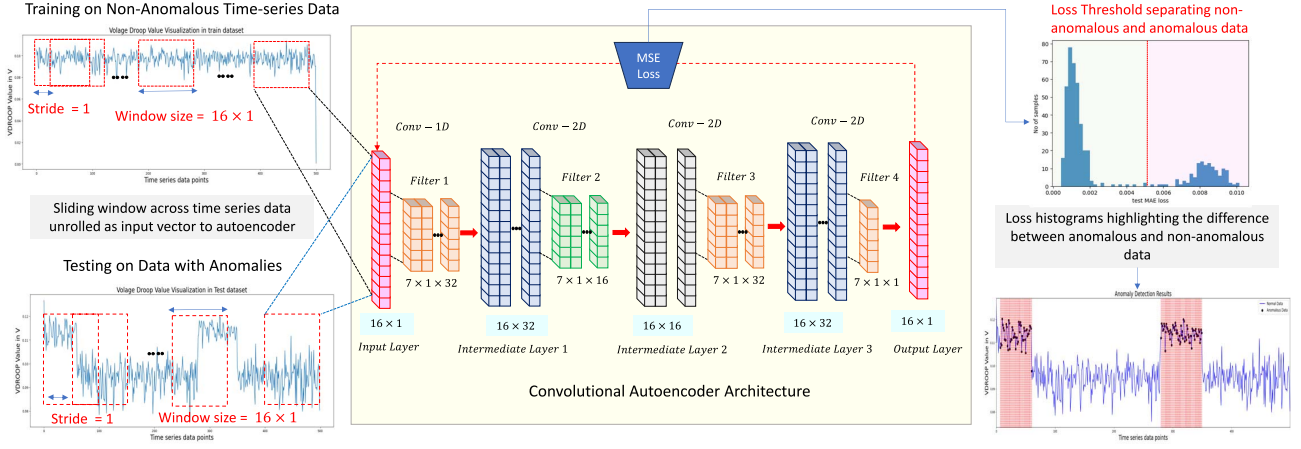


Fig. 10. Training and evaluation methodology for detecting anomalies in time series voltage droop data, including the sampling window for time series data and the architecture of the convolutional autoencoder.

Consider a time series sequence of l data samples given by $\Gamma = [\gamma_1, \gamma_2, \dots, \gamma_l]$. We can train autoencoders on time series data by batching the continuous sequence of input samples into groups based on a sliding window whose size is given by δ . Assuming that this moving window has a stride of s , the number of training samples is then given by $\lfloor (l - \delta) / s \rfloor$. The size of the input and output layers of the autoencoder is determined by the window size, δ .

B. Methodology for Detecting Voltage Droop Attacks

We develop a time series anomaly detection framework to detect voltage droop attacks in chiplet PDNs using convolutional autoencoders. The voltage droop values are sampled from standard cells that are part of timing critical paths in the chiplet design. The time series voltage droop data, $\mathcal{V}_X$, consists of 500 samples with the sampling window size, $\delta = 16$, and stride, $s = 1$. As a result, the size of the autoencoder's input and output layers is 16×1 . The autoencoder consists of three intermediate layers with filter sizes $7 \times 1 \times 32$, $7 \times 1 \times 16$, and $7 \times 1 \times 32$, respectively. Note that the padding after each layer is adjusted such that the $X - Y$ size of the intermediate feature maps remains the same at 7×1 . However, the feature depth is changed after every layer. The first intermediate layer upsamples the feature depth to 16×32 . The second intermediate layer downsamples the feature vector to a latent space of size 16×16 . This is then upsampled again by the third intermediate layer to create a feature map of size 16×32 . The final filter is responsible to shrink the output feature map to match the depth of the input feature at 16×1 . Fig. 10 illustrates the overall architecture of the convolutional autoencoder.

As described in Section V-C, time series voltage droop data for an IP is collected, with and without the activation of Trojans. The autoencoder is trained on nonanomalous voltage droop data, also known as in-distribution voltage droop data. Let the set $\mathcal{U}_{X,\text{train}}$ contain the reconstruction loss of the voltage droop samples in the training dataset for the IP X , $\mathcal{V}_{X,\text{train}}$. Therefore, $\mathcal{U}_{X,\text{train}} = [\mathcal{L}(\phi, \theta; V_1), \mathcal{L}(\phi, \theta; V_2), \dots, \mathcal{L}(\phi, \theta; V_N)]$, where N is the number of samples in the training dataset. The loss threshold

ϵ is set to the reconstruction loss of the sample with the maximum value in the set $\mathcal{U}_{X,\text{train}}$. Therefore, $\epsilon = \max\{\mathcal{U}_{X,\text{train}}\}$. This ensures that a time-series window during evaluation is flagged as anomalous only if the reconstruction loss for that sample is higher than the maximum loss obtained on the training samples. Furthermore, to reduce the number of false positives, a history-based approach is also utilized, where a voltage droop sample associated with the i th window, V_i , is not considered anomalous until $\mathcal{L}(\phi, \theta; V_i) > \epsilon$ and the loss of at least eight previous sample windows was also greater than ϵ , i.e., $\mathcal{L}(\phi, \theta; V_{i-j}) > \epsilon \forall j : (1 \leq j \leq 8)$, where j iterates through the previous eight samples leading up to the i th sample.

C. Results and Evaluation

We evaluate the effectiveness of the proposed autoencoder architecture on time series voltage droop data collected across different chiplet IPs from the CEP benchmark suite. Time series data is collected for the FIR, IIR, 3DES, and MD5 chiplet IPs across two different functional workload types, $\mathcal{W}_{X,\text{busy}}$ and $\mathcal{W}_{X,\text{bursty}}$. The autoencoder is trained on in-distribution data, i.e., data with Trojans not activated. For every IP, two anomaly detection models are trained, one each for the type of workload running on the IP. The training sets of voltage droop data, i.e., $\mathcal{V}_{X,\text{busy}}$ and $\mathcal{V}_{X,\text{bursty}}$, contain a sequence of 500 V_{droop} samples. The models are trained for 150 epochs with a learning rate of 0.001. Training on average takes less than 4 min for all the benchmarks. Training is done on an Intel Xeon Gold CPU running at 2.1 GHz. The trained models are then evaluated on a test set containing time series voltage droop data associated with the corresponding workloads with different number of Trojans activated. For both busy and bursty mode of workloads, four test sets are created for evaluation, with 15, 20, 25, and 30 Trojans activated at a time in each of the test set, respectively. The test set also consists of 500 voltage droop samples, with Trojans activated for 25% of the time and triggered randomly in the sequence.

The effectiveness of the convolutional autoencoder in detecting voltage droop anomalies stems from its ability to

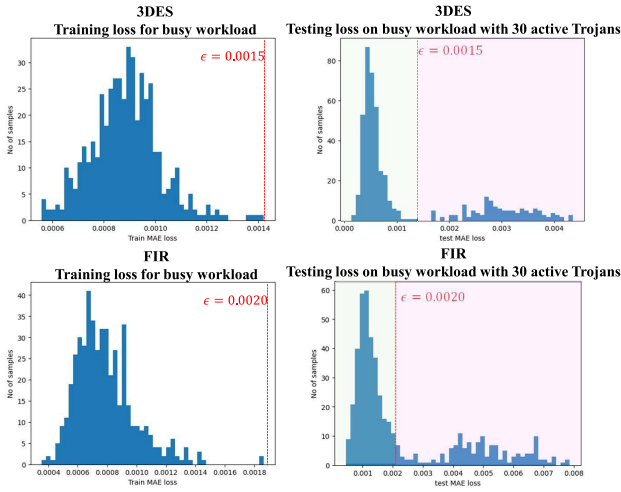


Fig. 11. Training (left) and testing (right) loss histograms for the 3DES and FIR chiplet IPs. Note that the loss threshold ϵ derived from the training loss histogram can be used to form a clear separation between anomalous data (in red) and nonanomalous data samples in the testing loss histograms.

TABLE III

PERFORMANCE OF CONVOLUTIONAL AUTOENCODER MODELS TRAINED ON BUSY AND BURSTY MODE WORKLOADS. TWO MODELS ARE TRAINED, ONE EACH TYPE OF WORKLOAD. THESE ARE EVALUATED ON TEST SETS CONTAINING DIFFERENT NUMBER OF ACTIVE TROJANS

IP	Eval.	Busy Workload ($V_{X,busy}$)				Bursty Workload ($V_{X,bursty}$)			
		Trojan Count				Trojan Count			
3DES	Acc.	15	20	25	30	15	20	25	30
	FP	98.4	98.2	98.4	91.4	97.4	95.6	93.63	95.22
	TE	0	0.4	0.2	6.8	0.2	0.6	4.77	1.81
IIR	Acc.	1.6	1.4	1.4	1.8	2.0	3.8	1.59	1.59
	FP	92.5	95.4	97.1	97.72	95.22	97.27	97.05	97.27
	TE	5.90	3.01	0.90	1.59	0	0.68	0.68	0.90
FIR	Acc.	1.59	1.59	2.04	0.68	4.77	2.04	2.27	1.81
	FP	97	98.63	98.16	97.95	92.24	93.54	95.68	89.77
	TE	1.0	0	1.83	1.36	5.94	5.32	0.22	9.54
MD5	Acc.	2.0	1.36	0	0.90	1.81	1.14	4.09	0.68
	FP	91.73	94.61	96.33	98.28	94.68	90.3	96.76	92.66
	TE	7.75	3.19	1.65	0.79	1.84	4.61	0	6.42
		0.5	2.19	2.01	0.91	3.46	4.84	3.23	0.91

Acc.: Accuracy of the model; FP: False Positives, i.e., non-anomalous samples that are classified as anomalous; TE: Test Escape, i.e., anomalous samples that are classified as non-anomalous.

distinguish between the distribution of an anomalous voltage droop sample from a nonanomalous one. This can be evaluated by distinguishing the reconstruction loss between a nonanomalous V_{droop} sample and an anomalous V_{droop} sample in the test set. To analyze this, we first plot the training loss histogram associated with the data samples in the training set. Fig. 11 shows the training loss histogram and the derived reconstruction loss threshold for both the FIR and 3DES chiplets running busy workloads. Note that the loss threshold, $\epsilon_{FIR} = 0.002$. Similarly, $\epsilon_{3DES} = 0.0015$. We observe that these reconstruction loss thresholds, when superimposed on the test loss histogram, are very effective in distinguishing between anomalous data points and nonanomalous data points. Observe that most of the outliers in case of both 3DES and FIR chiplets have a loss value much greater than their corresponding loss thresholds. This indicates that the convolutional autoencoder is effective in separating in-distribution data samples (nonanomalous data on which training took place) from out-of-distribution data samples (anomalous data samples).

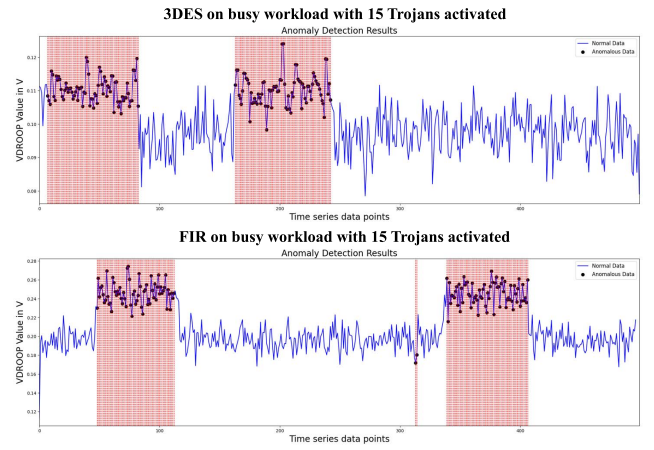


Fig. 12. Voltage droop anomalies detected by the convolutional autoencoder model in time series data.

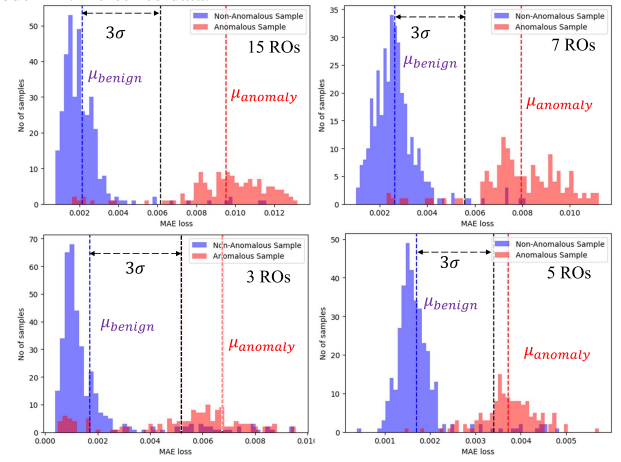


Fig. 13. Separation in the reconstruction loss distribution of anomalous and nonanomalous samples for the IIR chiplet with a variety of RO-based Trojans ranging from 3, 5, 7, and 15 ROs.

Table III shows the performance evaluation of the trained anomaly detection models on different test sets. For every IP, two models are trained, one each for the type of workload running on the IP. Similarly, for every workload, there exist four test sets that contain time series data with different number of Trojans activated at a time. For each test set, the set number of Trojans are randomly activated for 25% of the time. From Table III, it is clear that the convolutional autoencoder models achieve very high accuracy when evaluated on different number of activated Trojans across both busy and bursty mode workloads for all chiplet IPs. The false positives and test escape values are also less than 2%, with no false positives in some scenarios. Fig. 12 shows the voltage droop anomalies detected by the convolutional autoencoder model for the 3DES and FIR chiplets on the test set evaluating 15 active Trojans on a busy workload.

D. Intuitive Explanation of Autoencoder's Performance

The convolutional autoencoder is able to achieve good anomaly detection performance in dynamic operational settings. This is due to the model's ability to separate anomalous and nonanomalous distributions so well in the latent space. As a case study, we analyze the difference in the distribution of

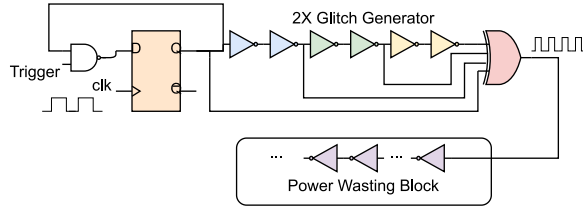


Fig. 14. Glitch-based power wasting hardware Trojan.

reconstruction loss for anomalous and nonanomalous samples for IIR chiplet with 15, 7, 5, and 3 active Trojans, respectively, while running a mixed workload, $\mathcal{W}_{\text{IIR,mix}}$. As shown in Fig. 13, we plot the mean of both the distributions (μ_{benign} , μ_{anomaly}) and observe a clear and significant separation between the two points. We next observe that the anomalous distribution is skewed significantly with μ_{anomaly} lying beyond the 3σ point from μ_{benign} , with less than 1% of anomalous samples lying within 3σ of μ_{benign} . Furthermore, we also evaluate the kurtosis score for both the distributions, which is the fourth standardized moment of a distribution and is given by $\mathcal{K}(X) = E((X - \mu)/\sigma)^4$. A higher-kurtosis score is indicative of a highly peaked distribution while a lower score is indicative of a greater spread. For IIR IP, $\mathcal{K}(X_{\text{benign},7}) = 11.86$ and $(X_{\text{anomaly},7}) = 1.62$. Similarly, $\mathcal{K}(X_{\text{benign},15}) = 22.10$ and $(X_{\text{anomaly},15}) = 2.87$. These values also suggest that the autoencoder can create highly peaked and narrow distributions for nonanomalous samples (higher $\mathcal{K}(X)$), thereby making the detection of anomalous samples that are well separated from these distributions easier (lower $\mathcal{K}(X)$). A similar trend is followed for lower Trojan counts with $\mathcal{K}(X_{\text{benign},5}) = 9.15$, $(X_{\text{anomaly},5}) = 0.75$, and $\mathcal{K}(X_{\text{benign},3}) = 7.05$ and $(X_{\text{anomaly},15}) = 0.43$.

VII. DISCUSSION

A. Time-Series Anomaly Detection on Non-RO Trojans

So far, we have analyzed the effectiveness of ROs in causing PDN voltage droop events leading to timing failure. ROs offer several advantages, including 1) high-switching activity; 2) ease of insertion and masking among large number of other ROs that are part of the design; and 3) relatively simply design. To investigate the effectiveness of the autoencoder-based time series anomaly detection model against non-RO Trojans, we implement a glitch-based power wasting Trojan as shown in Fig. 14. This Trojan is a variant of a power hammering circuit proposed in [42]. When the Trigger signal is enabled, a chain of six inverters is used to generate multiple variable delay glitches at the output of the XOR gate, which drives a power wasting block comprising of a chain of 10 additional inverters that propagate the glitch. The frequency of glitch generation is $N/2$ times that of the input clock, where N refers to the number of inputs to the XOR gate. In our implementation, $N = 4$, thereby producing two glitches every clock cycle. As compared to RO-based Trojan, the glitch-based Trojan generates almost twice the amount of switching activity and has an area footprint comparable to the RO-based Trojan.

We compare the performance of our time series anomaly detection model for both RO- and glitch-based Trojans. We

TABLE IV
PERFORMANCE OF THE CONVOLUTIONAL AUTOENCODER MODELS
TRAINED ON MIXED WORKLOADS FOR ALL BENCHMARK IPs
EVALUATED ON TEST SETS CONTAINING DIFFERENT NUMBER
OF RO- AND GLITCH-BASED TROJANS

IPs	Eval.	RO- based Trojan			Glitch-based Trojan		
		3	5	7	3	5	7
3DES	Acc.	68.32	90.3	92.6	74.3	85.8	90.78
	FP	31.2	5.3	4	22.8	12	7.12
	TE	1.48	4.4	3.4	2.9	1.2	0.6
IIR	Acc.	70.5	89.95	91.78	67.78	89.64	91.78
	FP	29.2	9.63	7.61	30.22	9.36	7.61
	TE	0.6	0.4	0.61	2	1	0.61
FIR	Acc.	72.5	91.2	93.4	75.6	88.8	92.6
	FP	16.5	4.8	6.6	22.2	8	6.8
	TE	11	4	0	2.2	3.2	0.6
MD5	Acc.	71.14	88.6	91.3	72.65	86.3	93.65
	FP	18.23	6.4	5.7	25.5	11.7	5.35
	TE	9.62	5	3	1.85	1.5	1
JPEG	Acc.	65.1	85.71	89.15	69.18	87	91.15
	FP	28.05	11.29	9.85	23.32	10.4	7.55
	TE	6.85	3	1	7.5	2.6	1.3

perform this analysis on IPs running on mixed workloads ($\mathcal{W}_{X,\text{mix}}$) as described in Section V-E. The convolutional autoencoder's Trojan detection performance is evaluated on three different test sets that contain both anomalous and nonanomalous time series data. From the RO-induced V_{droop} susceptibility assessment done earlier in Section V-E, we recall that some chiplets start exhibiting timing violations around the activation of five or more Trojans. Thus, to perform a practical assessment of the autoencoder performance on both Trojan designs, we activate three, five, and seven Trojans randomly over 25% of the time the three evaluation test sets, respectively. Table IV summarizes the convolutional autoencoder's performance across both Trojan types. We observe that the model performs significantly well in case of both Trojan types, achieving close to 90% anomaly detection accuracy in case of both five and seven Trojan insertions. Furthermore, we also observe that the model's test escape is significantly low even for three and five Trojan insertions. The high-false positive rate for three Trojans can be explained due to the increasing overlap of the tail distributions of both nonanomalous and anomalous distributions, which can also be seen in Fig. 13. It is important to note that as the Trojan count decreases, there is greater overlap in tail distributions of both anomalous and nonanomalous samples, with μ_{anomaly} shifting closer to the 3σ threshold of the nonanomalous distribution. This close proximity of an anomalous sample to a benign sample is also indicative of a lesser likelihood of timing path violation occurring for such scenarios. In fact, the convolutional autoencoder is effective at catching voltage droop anomalies that have a greater likelihood of causing timing violations due to their increased separation from the 3σ corner of the nonanomalous distribution, which is also why the model's test escape is low.

B. Mitigating PDN Voltage Droop Attacks

As discussed in Sections II and III, the nature of the threat model for HI assumes the use of untrusted COTS IP, which can lead to untrusted chiplets being integrated in the system. As RO-based Trojans are significantly small in size, they

can be easily incorporated into chiplets being fabricated and tested across third-party locations. Furthermore, due to their size and independence from control FSMs, such malicious modifications can easily sneak past existing security mechanisms that rely on mere visual inspection. Furthermore, even formal security verification techniques, such as logic equivalence checking, cannot guarantee security against Trojans that impact functional behavior through power-based side channels [43]. This necessitates the need for nonobtrusive design-level safe guards that can ensure the security of the overall HI system during runtime, especially against power-based side channel attacks. As voltage droop across a power rail is directly affected by the ability of that rail to supply power to its load, the higher the driving capacity of the rail, the lower is the voltage droop. We propose the use of power and ground pins (PG pins) to mitigate the impact of voltage droop-based attacks. The voltage droop distribution across a chiplet is determined by the dynamic loading conditions on the power and ground rails for the different instances (and/or regions) within the chiplet. The closer an instance is to a PG pin, the lower is the voltage droop experienced by that instance (and/or region) under dynamic loading conditions.

In order to prevent voltage droop-based PDN side-channel attacks, PG pins can thus be systematically inserted in locations that are: 1) experiencing high-voltage droop, also known as voltage droop hotspots and 2) susceptible to timing failure due to significantly low slack, i.e., have lower tolerance for voltage droop events. Inserting PG pins systematically across a chiplet not only reduces the susceptibility of specific paths to voltage droop-induced timing failure, but also reduces the worst-case value of voltage droop throughout the entire chiplet itself. Fig. 15 shows the worst-case voltage droop distribution for the FIR chiplet for a variety of PG pin configurations. Note that the worst-case voltage droop is maximum in Fig. 15(a), which has only one PG pin. Meanwhile, the worst-case voltage droop distribution is much less and more evenly spread out for the case with seven PG pins as shown in Fig. 15(d).

The methodology for inserting PG pins across chiplets can enhance security against voltage droop-based PDN timing fault attacks. This is because there are two limiting factors that affect timing degradation. V_{droop}^{max} : The maximum limit of the worst-case voltage droop. t_{slack}^{min} : The minimum slack threshold that must be required to meet timing requirements.

Given the list of paths $\mathcal{P}$ that must meet the timing requirement t_{slack}^{min} and the worst-case value V_{droop}^{max} that needs to be tolerated, we can compute the worst-case timing derating factors for the paths in $\mathcal{P}$. Based on this computation, we obtain a new list $\mathcal{P}_{derated}$, which contains a list of paths out of which some may now have become susceptible to timing failure (if $t_{slack}^{path} < t_{slack}^{min}$). We can now identify the instances belonging to these susceptible paths within the chiplet layout, and insert PG pins to power and ground rails belonging to those instances to prevent the adverse impact of voltage droop. This method provides designers the ability to guarantee timing security against voltage droop-based attacks given the worst-case voltage and timing thresholds. The impact of Trojans inserted in the active interposer and shared PDNs across

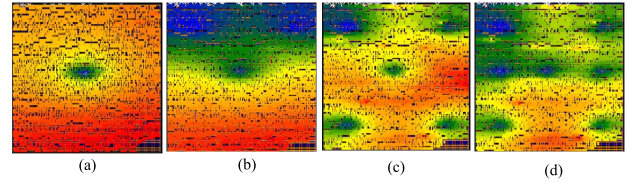


Fig. 15. Worst-case voltage droop across the FIR chiplet for different location and number of PG pins inserted in the PDN (blue-minimum at 0 V, red-maximum at 90 mV). (a) One PG pin at center; (b) three PG pins with one inserted at center and two in top half; (c) five PG pins across four corners and one in center; and (d) seven PG pins inserted with three across both left and right halves, and one in center.

multiple chiplets in HI systems can also be explored as part of future work.

VIII. CONCLUSION

We have presented the shortcomings associated with existing security solutions targeted toward runtime IP security and developed a methodology for the design, synthesis, layout, and simulation of voltage droop events for chiplet IPs. We have described a method to analyze the susceptibility of chiplet IPs toward voltage droop-based PDN timing attacks, resulting in timing failure during functional runtime. We have developed a convolutional autoencoder-based method that can detect voltage droop anomalies in time series data, caused by RO-based hardware Trojans in chiplet PDNs. This method achieves very high accuracy with minimal amount of false positives and test escape.

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